



Google's shot at wearable computing

Google Glass is a computer-integrated pair of glasses that realises augmented reality. That is, it supplements what you see with relevant information. If you are walking down a street, it could pop up information on the buildings there, notify you of a traffic diversion or suggest a restaurant – somewhat like today's location-aware phones can. It is likely to be launched around 2014, but has already gained a lot of attention.

Inside: Glass is likely to include a heads-up display; standalone processor; on-board memory and battery; projection technique such as the Lumus optical engine module; a 3.2MP camera and 1080p video recording; touch pad, microphone and speakers for input-output; sensors like gyroscope, accelerometer and magnetometer (e-compass) for context-sensitivity and gesture-enabled interaction; GPS functionality; and 3G/4G, Wi-Fi and Bluetooth connectivity. It is proposed to run on Android. Although the early prototype uses an opaque screen, the actual might use a transparent liquid-crystal display or active-matrix organic light-emitting diode display.

NASA's best Martian bet

NASA's Curiosity is the largest rover sent to Mars. The car-sized rover apparently contains much more scientific instrumentation than other rovers sent to the Red Planet. According to NASA's Mars Lab, the rover has everything that a living thing needs to breathe, live and work.

Inside: The rover's body is a structure that protects the stuff inside. Computers are its brain, and temperature is controlled through internal heaters, a layer of insulation and more. The rover's neck and head serve as a mast for the cameras to give the rover a human-scale view; the cameras and instruments are its eyes and senses. It has robotic arms and hands to collect material samples for study, wheels and legs for mobility, batteries for energy, and ultra-high frequency antennae to communicate! One of the two computers inside the rover works all the time, while the other serves as backup. Both use radiation-hardened BAE RAD750 microchips operating at up to 200 MHz. Each computer is equipped with 2GB flash memory, 256MB random-access memory and 256kB erasable programmable read-only memory. While this configuration seems much lower than today's computers, these chips are used because they are radiation hardened, through a very high-tech and costly process to stand the rigours of a strange planet. ●



The author is a technically-qualified freelance writer, editor and hands-on mom based in Chennai